

CodeWeaver comparison with the complete CRUST-Bench citation corpus

We reconciled all 30 Semantic Scholar citation records for CRUST-Bench into 19 unique migration-relevant works, eight tangential works, one out-of-scope work, and two duplicate records. Every in-scope work and every empirical surface found in its paper is inventoried. ORBIT's exact 24-project subset is newly evaluated from the frozen CodeWeaver three-repetition campaign, and ACToR's six-program absolute experiment is newly evaluated with a qualified 492-case hidden differential oracle over 18 CodeWeaver cells. Schesch/Ernst ACTOR's paper-87 values are separated from the pinned public artifact's 95 project directories. Other scores are reused only where subject identity and fixed contracts were already verified. Unreleased or metric-incompatible studies remain explicit blockers, never synthetic zeros or inferred wins.

Census method

The primary citation graph was Semantic Scholar's complete 30-record edge list as of 2026-08-20, cross-checked against OpenAlex, arXiv, publisher pages, and artifact repositories. InariRoll was resolved as a Hayroll duplicate, and two title variants were resolved as one Schesch/Ernst ACTOR paper.

Measurement rule

A CodeWeaver score is included only when the exact retained subject and a fixed independent oracle are available. Shared names without revision/hash equality, function-level validation metrics, interface inference, annotation synthesis, and terminal-agent task accuracy are not relabeled as repository pass-all.

ORBIT result

ORBIT reports 22/24 test-successful projects in both interface modes. Across the same 24 named projects, CodeWeaver's three repetitions average 45.83% project pass-all. ORBIT is an apparent single run while CodeWeaver retains all three outcomes, so this is an exact-subject descriptive comparison rather than a controlled architecture ablation.

ACToR absolute micro result

The pinned Li et al. artifact exposes six micro utilities, 15 seed tests per utility, and a separate fixed validation suite of 70, 100, 66, 84, 83, and 89 cases (492 total). Each C reference passed its own full contract before the validation files were kept outside all model-readable workspaces. Across 18 CodeWeaver cells, 14/18 passed every hidden case and 14/18 also contained no candidate-owned `unsafe` token or process delegation. Candidate binaries ran in a mount/PID-namespace chroot where reference and contract contents were masked by read-only empty mounts, Linux capabilities were cleared, and system executables were absent. All six compiling stub negative controls failed the full contract (1/492 cases passed), confirming that pass-all is non-vacuous. The public fixed contracts are included in the result package only after execution, with per-file checksums. A fully pre-model unauthenticated launcher attempt is archived as infrastructure evidence and excluded from all 18 measured cells. The shared fixed oracle supports a same-subject descriptive comparison, not a controlled ACToR architecture ablation: CodeWeaver uses its frozen five-repair, three-parity protocol, whereas the paper reports naive, collaborative, and ten-iteration ACToR configurations.

ACTOR denominator boundary

Schesch and Ernst report Figure 6 over 87 CRUST-Bench projects, but the pinned public results submodule contains 95 CRUST project directories and does not encode the paper's 13-project exclusion set. The paper's 87-denominator values therefore remain exact references only. CodeWeaver outcomes over the independently verifiable 95-directory

overlap are a separate table and are never labeled as a reproduction of Figure 6.

Availability boundary

RustPrint, DepTrans, PtrTrans, RustAssure, and several adjacent systems do not expose the exact evaluated revisions and fixed contracts needed for leakage-safe execution. Li et al. ACToR's 57-program macro metric remains blocked because it is cross-testing against unreleased system-generated outputs/tests, while its six-program absolute experiment is measured here. Schesch/Ernst ACTOR's exact paper-87 membership remains unresolved. EvoC2Rust's C2R-Bench/industrial set and DepTrans's Huawei set are explicitly unreleased. The package records these blockers and the maximum public comparison for each paper.

Prior-result audit

The five earlier paper packages now include a complete source-paper surface inventory, structured reference tables for previously omitted dataset/error/cost/ablation results, CodeWeaver cost and coverage telemetry, corrected version labels, and final-output Clippy analysis where executable.

Per-paper publication profiles

The `paper-profiles/` directory contains one human-readable PDF, Markdown report, complete empirical-surface inventory, decision record, and available structured reference tables for each of the 20 included works. These profiles link compatible standalone CodeWeaver packages and retain blockers for incompatible surfaces.

Citation reconciliation

Population	Count	Disposition
Semantic Scholar records	30	19 unique in-scope + 8 tangential + 1 out-of-scope + 2 duplicates
Unique in-scope works	19	all included in the decision matrix
Tangential works	8	TACO retained because it evaluates CRUST-Bench
Duplicate index records	2	resolved to Hayroll and Schesch/Ernst ACTOR

Exact ORBIT 24-project comparison

System	Build	Pass all	Fixed tests	Protocol
ORBIT expert interfaces	24/24	22/24 (91.67%)	not reported	paper reference; apparent single run
ORBIT generated interfaces	24/24	22/24 (91.67%)	not reported	paper reference; apparent single run
CodeWeaver repetition 1	24/24	11/24 (45.83%)	97/188	CodeWeaver multi-stage, 5 repairs, 3 parity rounds
CodeWeaver repetition 2	24/24	11/24 (45.83%)	95/188	CodeWeaver multi-stage, 5 repairs, 3 parity rounds

System	Build	Pass all	Fixed tests	Protocol
CodeWeaver repetition 3	24/24	11/24 (45.83%)	79/188	CodeWeaver multi-stage, 5 repairs, 3 parity rounds

ORBIT per-subject outcomes

Project	LoC	ORBIT expert	ORBIT generated	CW passing reps	CW fixed tests
CircularBuffer	213	pass	pass	3/3	3/3
Simple-Config	719	pass	pass	3/3	21/21
VaultSync	1121	pass	pass	0/3	0/3
mvptree	1121	pass	pass	2/3	2/3
razz_simulation	1145	pass	pass	0/3	0/3
Remimu	1162	pass	pass	3/3	3/3
libpgn	1162	fail	pass	0/3	50/60
LTRE	1212	pass	fail	2/3	2/3
lambda-calculus-eval	1264	pass	pass	2/3	65/66
jccc	1310	pass	pass	3/3	12/12
libpsbt	1331	pass	pass	1/3	10/12
mdb	1340	pass	pass	3/3	12/12
Genetic-neural-network-for-simple-control	1367	pass	pass	0/3	0/69
impcheck	1397	fail	pass	0/3	0/18
cfsm	1446	pass	pass	0/3	0/3
worsp	1494	pass	pass	3/3	12/12
libutf	1525	pass	pass	0/3	0/33
kairoCompiler	1589	pass	pass	0/3	0/6
fslib	1958	pass	pass	0/3	38/162
XOpt	2113	pass	pass	3/3	6/6
recordManager	2400	pass	pass	3/3	9/9
libm17	2901	pass	pass	0/3	12/21
tisp	3562	pass	fail	0/3	0/3
Megalania	3621	pass	pass	2/3	14/21

Existing publication-ready CodeWeaver evidence

Study	Scope	Headline	Result path
ReCodeAgent	118 projects / four language pairs	full raw data, baselines, ablations, tables, figures, and PDF	results/recodeagent-gpt-5.6-sol-final-2026-08-11
CRUST-Bench	100 projects x 3 repetitions	300/300 build; 165/300 pass all	results/crust-bench-codewea-ver-comparison-2026-08-14
SACTOR exact subset	50 projects x 3 repetitions	150/150 build; 92/150 pass all	results/sactor-codeweaver-comparison-2026-08-14
Rustine	23 projects x 1 repetition	21/23 compile; 10/21 fixed-contract pass	results/rustine-codeweaver-comparison-2026-08-12
EvoC2Rust public Vivo-Bench	15 groups / 19 modules x 3 repetitions	100% incremental compilation, fill compilation, and fixed-test rate; mean SafeRate 30.92%	results/evoc2rust-codeweaver-comparison-2026-08-13
RepoTransBench historical slice	3 projects x 3 repetitions	9/9 independently build and pass all fixed tests	results/repotransbench-code-weaver-comparison-2026-08-14
RustRepoTrans language slice	3 tasks x 3 repetitions	9/9 independently build and pass all fixed tests	results/rustrepotrans-codewea-ver-comparison-2026-08-14

Schesch/Ernst ACTOR public-artifact 95-project overlap

System	Build	Pass all	Fixed tests	Boundary
CodeWeaver repetition 1	95/95	56/95	425/564	unresolved; not treated as the paper's 87-project slice
CodeWeaver repetition 2	95/95	55/95	433/564	unresolved; not treated as the paper's 87-project slice
CodeWeaver repetition 3	95/95	54/95	403/564	unresolved; not treated as the paper's 87-project slice

Li et al. ACToR six-program absolute hidden-oracle result

Subject	Build	Pass all	Safe pass	Hidden tests	Test rate
csplit	3/3	2/3	2/3	209/210	99.52%
expr	3/3	2/3	2/3	200/300	66.67%
fmt	3/3	3/3	3/3	198/198	100.00%
join	3/3	2/3	2/3	250/252	99.21%
printf	3/3	2/3	2/3	238/249	95.58%
test	3/3	3/3	3/3	267/267	100.00%
ALL	18/18	14/18	14/18	1362/1476	92.28%

Li et al. ACToR CodeWeaver execution telemetry

Subject	Elapsed hours	Output tokens	AIU	Premium requests
csplit	4.17	unavailable (unavailable)	4909.659	49
expr	3.53	unavailable (unavailable)	4231.305	61
fmt	3.62	unavailable (unavailable)	2542.231	53
join	4.17	unavailable (unavailable)	3105.143	59
printf	4.17	unavailable (unavailable)	2609.613	55
test	3.36	unavailable (unavailable)	2367.341	60
ALL	23.02	unavailable (unavailable)	19765.291	337

Li et al. ACToR published reference results

Surface	Scope	System	Metric	Value	Uncertainty
micro evaluation	6 utilities / 3 runs	ACToR Claude Code Sonnet 4.5	hidden-test pass percent	97.0	SD 1.9 pp
micro evaluation	6 utilities	naive Claude Code Sonnet 4.5	hidden-test pass percent	89.2	not reported
micro evaluation	6 utilities / 10 iterations	ACToR Claude Code Sonnet 4.5	hidden-test pass percent	98.2	not reported
macro evaluation	57 BSD utilities	coverage baseline	relative pass percent	58.4	not reported
macro evaluation	57 BSD utilities	ACToR	relative pass percent	95.1	not reported
C2SaferRust augmentation	7 executables	C2SaferRust	pass percent	76.3	not reported
C2SaferRust augmentation	7 executables	C2SaferRust + ACToR	pass percent	92.9	not reported
cost	57 BSD utilities	coverage baseline	USD	808	not reported
cost	57 BSD utilities	ACToR	USD	1634	not reported

Complete inclusion and execution matrix

Paper	Empirical scope	Status	Reason
ORBIT	24 named CRUST-Bench projects plus TRACTOR	measured_existing_slice	all 24 names map exactly to the retained 100-project campaign

Paper	Empirical scope	Status	Reason
SACTOR	TransCoder-IR, CodeNet, 50 CRUST-Bench projects, libogg	exact_existing_result	three CodeWeaver repetitions on the exact 50-project subset
Rustine	23 C repositories	exact_existing_result	leakage-safe CodeWeaver evaluation on all 23 released subjects
EvoC2Rust	Vivo-Bench, C2R-Bench, six industrial projects	partial_existing_result	three repetitions cover public Vivo-Bench; C2R-Bench and industrial projects are unreleased
RustPrint	eight repositories, 11K-84K LoC	public_artifact_new_run_required	CodeWikiBench's eight repositories are disjoint from retained CRUST-Bench tasks
ReCodeAgent	118 repositories, four language pairs	exact_existing_result	complete 118-project CodeWeaver campaign and paper-style report
DepTrans	145-instance benchmark and 15 Huawei projects	blocked	no exact public benchmark source and fixed oracle
PtrTrans	16 Crown projects	public_artifact_license_restricted	Crown-16 is disjoint from CRUST-Bench and cannot be redistributed without a license
Reininator;	C-to-Rust interface translation	not_comparable	interface synthesis is not end-to-end project translation
RustAssure	five applications/libraries; differential symbolic validation	public_artifact_metric_mismatch	function-level KLEE equivalence is not CodeWeaver project pass-all
ACToR	63 command-line C utilities	exact_micro_campaign_and_blocked_macro	the six-program absolute hidden oracle is executable and receives three leakage-safe CodeWeaver repetitions; the 57-program macro metric is relative cross-testing without released fixed outputs/tests
MatchFixAgent	translation validation and repair	not_comparable	validator/repair verdict metrics are not translator success metrics
LAC2R	GNU coreutils, Laertes, TRACTOR public tests	reference_only	eight subject names overlap the Rustine suite, but revisions/contracts are not proven identical
Fine-Tuning Qwen3-72B	SACTOR-framework function translation	not_comparable	function-level SACTOR evaluation is not a repository project oracle

Paper	Empirical scope	Status	Reason
Hayroll	macro and conditional-compilation wrapper translation	not_comparable	specialized preprocessing task has no shared end-to-end metric
CNnotator	memory-safety annotation synthesis	not_comparable	annotation synthesis is not repository translation
TRAM	mock-based Java-to-Python in-isolation validation	not_comparable	validation method and language pair differ
Formal Compositional Reasoning	formal compositional code-translation reasoning	not_comparable	methodology/position contribution lacks a compatible released project oracle
ACTOR (Schesch and Ernst)	paper: 87 CRUST-Bench projects and 338 TRACTOR cases; public result artifact: 95 CRUST project directories	reference_87_and_measure_d_public_95_overlap	paper values retain denominator 87, but the pinned results submodule contains 95 project directories and does not identify the paper's 13 exclusions; CodeWeaver's matching public 95 are reported separately
TACO	terminal-agent context compression over six benchmarks	reference_only	47.00% to 48.05% terminal-task accuracy is not project pass-all

ACTOR CRUST-Bench reference (Figure 6)

System	Setting	Build	Pass all	LoC	Unsafe
ACTOR Kiro	no benchmark tests	82/87	56/87	43000	1%
C2Rust/Laertes/C2SaferRust/SmartC2Rust	no benchmark tests	0/87	0/87	not reported	not reported
GPT-5.4	no benchmark tests	82/87	50/87	57000	0%
Kimi K2.5	no benchmark tests	46/87	31/87	28000	0%
Gemini 3.1 Pro	no benchmark tests	11/87	8/87	15000	0%
ACTOR Kiro	test repair	87/87	82/87	52000	1%
ACTOR Claude	test repair	85/87	75/87	45000	0%
ACTOR Codex	test repair	87/87	81/87	48000	1%
GPT-5.4	test repair	79/87	64/87	58000	0%
Kimi K2.5	test repair	45/87	39/87	28000	0%
Gemini 3.1 Pro	test repair	8/87	7/87	15000	0%

Public-artifact reference highlights

Paper	Scope	System	Metric	Value
RustPrint	8	RustPrint Kimi	compiled repositories	8
RustPrint	8	RustPrint GPT-5.4	compiled repositories	8
RustPrint	8	Self-Repair	compiled repositories	0
RustPrint	8	EvoC2Rust	compiled repositories	0
RustPrint	8	C2Rust	compiled repositories	8
RustPrint	8	Claude Code	compiled repositories	8
RustPrint	16 cells	RustPrint GPT-5.4	aggregate cross-test TPR percent	98.7
RustPrint	16 cells	RustPrint Kimi	aggregate cross-test TPR percent	95.17
RustPrint	16 cells	Claude Code	aggregate cross-test TPR percent	79.85
RustPrint	8 repositories	RustPrint Kimi	feature conservation percent	93.26
RustPrint	8 repositories	RustPrint GPT-5.4	feature conservation percent	97.76
RustPrint	8 repositories	Claude Code Kimi	feature conservation percent	52.52
RustPrint	8 repositories	Claude Code GPT-5.4	feature conservation percent	48.87
RustPrint	8 repositories	RustPrint Kimi	SafeRate A/F percent	96.23/96.19
RustPrint	8 repositories	RustPrint GPT-5.4	SafeRate A/F percent	99.41/98.47
PtrTrans	Crown-16 total	Crown	lint alerts	6802
PtrTrans	Crown-16 total	PR2	lint alerts	4135
PtrTrans	Crown-16 total	PtrTrans	lint alerts	349
PtrTrans	Crown-16 total	Crown	unsafe usages	141866
PtrTrans	Crown-16 total	PR2	unsafe usages	134185
PtrTrans	Crown-16 total	PtrTrans	unsafe usages	85
PtrTrans	small projects	FLOURINE	compiled/equivalent percent	69.9/52.3
PtrTrans	small projects	PtrTrans	compiled/equivalent percent	98.3/81.6
PtrTrans	large projects	FLOURINE	compiled/equivalent percent	64.0/14.2
PtrTrans	large projects	PtrTrans	compiled/equivalent percent	85.9/67.9
PtrTrans	small-10 average	PtrTrans_PS	compiled/equivalent percent	89.3/59.5
PtrTrans	small-10 average	PtrTrans_PU	compiled/equivalent percent	84.6/52.9

Paper	Scope	System	Metric	Value
PtrTrans	small-10 average	PtrTrans_RA	compiled/equivalent percent	87.9/61.9
PtrTrans	small-10 average	PtrTrans_EC	compiled/equivalent percent	66.0/50.8
PtrTrans	small-10 average	PtrTrans	compiled/equivalent percent	100/81.6